

Reading a Rational Graph from Its Formula

Answer Key

Algebra 2 Rational Functions

Quick Answers

1. $(x - 3)(x + 3), (x - 3)(x + 5)$

2. $\frac{x-3}{x-2}, \frac{5}{4}$

3. 5, -2

4. 4, -2, $-\frac{8}{3}$

5. 3

6. $\frac{1}{2}, 3$

7. 4, -4, $-\frac{1}{8}$

8. 0

9. 2, -1, $\frac{3}{4}$

10. 4

11. 2, -6

12. $\frac{8}{3}, 4, 2$

Curriculum

Level: Algebra 2 Rational Functions

HSA-APR.D.6–D.7 – *problems 1, 2, 12*

HSF-IF.A–HSF-IF.C – *problems 3, 4, 5, 6, 7, 8, 9, 10, 11*

Difficulty 1–5 of 5 across 23 tagged checks.

Common wrong answers

6. *If they got -0.5: divided -6 by 12 instead of by -12, so the answer came out negative*

1. Factor $f(x) = \frac{x^2 - 9}{x^2 + 2x - 15}$ **completely.**

Step 1. Numerator: $x^2 - 9$ is a difference of squares, so it factors as $(x - 3)(x + 3)$.

Step 2. Denominator: find two numbers with product -15 and sum $+2$. Those are $+5$ and -3 , so $x^2 + 2x - 15 = (x + 5)(x - 3)$.

Step 3. Write the fraction with both factorisations in place.

$f(x) = \frac{(x - 3)(x + 3)}{(x + 5)(x - 3)}$
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Why this matters: the shared factor $(x - 3)$ is already visible, and it is the reason $x = 3$ will be a hole rather than an asymptote in later problems.

2. $g(x) = \frac{x^2 - x - 6}{x^2 - 4}$: lowest terms, and the hole.

Step 1. Factor both parts: $\frac{(x-3)(x+2)}{(x-2)(x+2)}$.

Step 2. The factor $(x+2)$ appears top and bottom, so it cancels for every x except $x = -2$, where the original g is undefined.

$$g(x) = \frac{x-3}{x-2}$$

Step 3. The height of the hole is the value the simplified form takes at $x = -2$: $\frac{-2-3}{-2-2} =$

$$\frac{-5}{-4} = \frac{5}{4}.$$

$$\text{hole at } \left(-2, \frac{5}{4}\right)$$

Grading note: the simplified formula alone is only half the answer — the excluded value $x = -2$ must be stated, because g and $\frac{x-3}{x-2}$ are not the same function.

3. Vertical asymptotes of $h(x) = \frac{x+1}{x^2 - 3x - 10}$.

Step 1. Factor the denominator: two numbers with product -10 and sum -3 are -5 and $+2$, so $x^2 - 3x - 10 = (x-5)(x+2)$.

Step 2. The numerator $x+1$ is not $(x-5)$ or $(x+2)$, so nothing cancels — every zero of the denominator survives as a vertical asymptote.

Step 3. Set each surviving factor to zero: $x-5=0$ and $x+2=0$.

$$x = 5$$

$$x = -2$$

Listen for: a student who names $x = -1$ as an asymptote. That is the zero of the NUMERATOR, which is an x -intercept, not an asymptote.

4. Intercepts of $p(x) = \frac{x^2 - 2x - 8}{x^2 + 4x + 3}$.

Step 1. x -intercepts come from the numerator: factor $x^2 - 2x - 8 = (x-4)(x+2)$, so the numerator is zero at $x = 4$ and $x = -2$.

Step 2. Check that neither value breaks the denominator: $x^2 + 4x + 3 = (x+1)(x+3)$ is zero only at $x = -1$ and $x = -3$. Both candidates survive.

$$x = 4$$

$$x = -2$$

Step 3. The y -intercept is $p(0) = \frac{0-0-8}{0+0+3} = \frac{-8}{3}$.

$$\left(0, -\frac{8}{3}\right)$$

Grading note: $-\frac{8}{3} \approx -2.67$; a decimal answer is fine if the fraction is shown first.

5. Horizontal asymptote of $q(x) = \frac{3x^2 - 5}{x^2 + 4}$.

Step 1. Numerator and denominator both have degree 2, so neither grows faster — the graph settles onto the ratio of the leading coefficients rather than onto $y = 0$.

Step 2. Divide top and bottom by x^2 : $\frac{3 - 5/x^2}{1 + 4/x^2}$. As x grows without bound, $5/x^2 \rightarrow 0$ and $4/x^2 \rightarrow 0$, leaving $\frac{3}{1}$.

$$\boxed{y = 3}$$

Teaching note: the graph is allowed to sit below the line for small x — a horizontal asymptote describes the far left and far right only.

6. Dana's intercepts of $r(x) = \frac{2x - 6}{x^2 - x - 12}$.

Step 1. Recompute $r(0)$ carefully. Numerator at $x = 0$: -6 . Denominator at $x = 0$: -12 . The quotient of two negatives is positive.

Step 2. $r(0) = \frac{-6}{-12} = \frac{1}{2}$, so Dana's mistake was dropping the denominator's minus sign.

$$\boxed{(0, \frac{1}{2})}$$

Step 3. For the x -intercept, set the numerator to zero: $2x - 6 = 0$, so $x = 3$. The denominator at $x = 3$ is $9 - 3 - 12 = -6 \neq 0$, so the point really is on the graph.

$$\boxed{(3, 0)}$$

What Dana did wrong: she divided -6 by 12 instead of by -12 , which flipped the sign of a correct-looking answer.

7. $s(x) = \frac{x + 4}{x^2 - 16}$: asymptote or hole?

Step 1. s is undefined wherever the denominator is zero: $x^2 - 16 = (x - 4)(x + 4) = 0$ gives $x = 4$ and $x = -4$.

$$\boxed{x = 4}$$

$$\boxed{x = -4}$$

Step 2. Cancel: $\frac{x + 4}{(x - 4)(x + 4)} = \frac{1}{x - 4}$ for $x \neq -4$. The factor $(x - 4)$ is still in the denominator, so $x = 4$ is the vertical asymptote; $(x + 4)$ cancelled, so $x = -4$ is a hole.

Step 3. The height of the hole is the simplified value at $x = -4$: $\frac{1}{-4 - 4} = -\frac{1}{8}$.

$$\boxed{\text{hole at } (-4, -\frac{1}{8})}$$

Listen for: “two asymptotes” — the standard error. Cancelling first is what tells the two cases apart.

8. Horizontal asymptote of $t(x) = \frac{2x+7}{x^2+3x}$.

Step 1. The denominator has degree 2 and the numerator degree 1, so the bottom grows faster; a fraction with a fixed-size top and a runaway bottom is heading for zero.

Step 2. Divide through by x^2 : $\frac{2/x + 7/x^2}{1 + 3/x}$. Every term with an x underneath tends to 0, so the whole quotient tends to 0.

$$y = 0$$

In words: far to the right the graph flattens onto the x -axis from above, getting arbitrarily close without reaching it.

9. $u(x) = \frac{x^2 - x - 2}{x^2 - 4}$: which numerator zero is an intercept?

Step 1. Factor everything: $\frac{(x-2)(x+1)}{(x-2)(x+2)}$. The numerator is zero at $x = 2$ and $x = -1$.

$$x = 2$$

$$x = -1$$

Step 2. $(x-2)$ cancels, so $x = 2$ is not in the domain of u at all — a point cannot be an intercept if the function has no value there. Only $x = -1$ is an x -intercept.

Step 3. At the cancelled value, the graph has a hole whose height is the simplified value there: $\frac{2+1}{2+2} = \frac{3}{4}$.

$$\text{hole at } (2, \frac{3}{4})$$

Full answer: x -intercept at $(-1, 0)$; hole at $(2, \frac{3}{4})$.

10. Jonah's claim about $v(x) = \frac{x^2 - 4}{x - 2}$.

Step 1. Jonah's rule — “denominator zero, therefore vertical asymptote” — skips the factoring step. Factor first: $\frac{(x-2)(x+2)}{x-2}$.

Step 2. The $(x-2)$ cancels, leaving $v(x) = x+2$ for every $x \neq 2$. Nothing blows up: as x approaches 2 from either side, $v(x)$ approaches $2+2=4$.

$$\text{hole at } (2, 4)$$

Step 3. *Why he is wrong (model answer):* a vertical asymptote needs the function to grow without bound near the value, which happens only when a factor is left in the denominator after cancelling. Here the factor cancels, so v agrees with the line $y = x+2$ everywhere except at $x = 2$, where it simply has no value — an open circle at $(2, 4)$.

Manual review: the explanation is the graded part; the boxed point is the check.

11. $w(x) = \frac{2x^2 + 3x}{x^2 - 9}$: asymptote, and crossing it.

Step 1. Equal degrees again, so divide by x^2 : $\frac{2 + 3/x}{1 - 9/x^2} \rightarrow \frac{2}{1}$ as x grows.

$$y = 2$$

Step 2. A crossing means $w(x) = 2$, so solve $\frac{2x^2 + 3x}{x^2 - 9} = 2$, i.e. $2x^2 + 3x = 2(x^2 - 9)$.

Step 3. The $2x^2$ cancels: $3x = -18$, so $x = -6$. Check it is in the domain: $(-6)^2 - 9 = 27 \neq 0$, and $w(-6) = \frac{72 - 18}{27} = 2 \checkmark$

$$x = -6$$

Teaching note: a horizontal asymptote constrains the far ends of a graph, not the middle — crossing it once is perfectly normal.

12. Challenge: full analysis of $F(x) = \frac{2x^2 - 8}{x^2 - x - 2}$.

Step 1. Factor: $\frac{2(x-2)(x+2)}{(x-2)(x+1)}$, so $F(x) = \frac{2(x+2)}{x+1}$ for $x \neq 2$.

Step 2. The cancelled factor $(x-2)$ gives a hole at $x = 2$, height $\frac{2(2+2)}{2+1} = \frac{8}{3}$.

$$\text{hole at } (2, \frac{8}{3})$$

Step 3. The surviving factor $(x+1)$ gives the vertical asymptote $x = -1$; the simplified numerator is zero at $x = -2$, which is the x -intercept.

$$x = -1$$

$$(-2, 0)$$

Step 4. y -intercept: $F(0) = \frac{0 - 8}{0 - 0 - 2} = \frac{-8}{-2} = 4$.

$$(0, 4)$$

Step 5. Degrees are equal in the original formula, so the horizontal asymptote is the ratio of leading coefficients, $\frac{2}{1}$.

$$y = 2$$

Model sketch (manual review): two branches of $y = \frac{2(x+2)}{x+1}$ separated by the dashed line $x = -1$, both approaching the dashed line $y = 2$ far left and far right, crossing the axes at $(-2, 0)$ and $(0, 4)$, with an open circle at $(2, \frac{8}{3})$.